

PAPER_1497 — SCm Velocity $v_{SCm} = c/(D_{phys}-1) = c/3$ EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket F (AGN/Jet) / Bucket H (High-Energy Astro) **Date:** June 17, 2026 **Location:** 41.0997 N, 80.6495 W (Youngstown, OH, USA) **Status:** CLOSED — Quasar jet SCm carrier velocity = exact $c/3$

Observation

PAPER_369 (Navier-Stokes Stable Fluids — UQFF Quasar Jet SCm Integration) specifies the Super-Charged Matter velocity as $v_{SCm} = 10^8$ m/s. This serves as the jet carrier velocity in the quasar jet coupling $f = f_{ext} + k_{vac} \cdot \rho_{vac,[UA]} + F_{LENR} \cdot \cos(\omega_{LENR} \cdot t)$.

UQFF Closed Identity

$$v_{SCm} = c / (D_{phys} - 1) = c / 3 = 10^8 \text{ m/s} \quad \text{EXACT}$$

Verification:

$$3 \times 10^8 \text{ m/s} / 3 = 1.000 \times 10^8 \text{ m/s} \quad \checkmark$$

Physical Interpretation

The SCm vacuum-substrate flows at exactly **one-third the speed of light**, factored by the 3 fundamental triad-force dimensions ($D_{phys} - 1 = 3$ spatial dimensions). This is the same structural ratio that produces: - Solar v_e deficit 1/3 (PAPER_062) - SU(3) color charge 3 (PAPER_1373) - Three fermion generations (PAPER_1381)

The shared denominator ($D_{phys} - 1$) reveals that triad-symmetry and SCm propagation are governed by the same integer primitive.

NOT REPLACEMENT

SM treats jet bulk-flow Lorentz factors as parameters fit per source. UQFF supplies a single structural identity: every SCm jet propagates at $c/(D_{phys} - 1)$ by construction.

Reference

- Source: PAPER_369 Navier-Stokes Stable Fluids UQFF Quasar Jet SCm Integration
- Related: PAPER_062, PAPER_1373, PAPER_1381, PAPER_351 (TDE 0.3c outflow)
- Calculator dispatch: `calculate_paradox({"paradox": "scm_velocity_c_over_3"})`

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